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United States Patent	12399008
Kind Code	B2
Date of Patent	August 26, 2025
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Fused quartz dual shell resonator and method of fabrication

Abstract

A dual-shell architecture and methods of fabrication of fused quartz resonators is disclosed. The architecture may include two encapsulated and concentric cavities using plasma-activated wafer bonding followed by the high-temperature glassblowing. The dual-shell architecture can provide a protective shield as well as a “fixed-fixed” anchor for the sensing element of the resonators. Structures can be instrumented to operate as a resonator, a gyroscope, or other vibratory sensor and for precision operation in a harsh environment. Methods for fabricating a dual-shell resonator structure can include pre-etching cavities on a cap wafer, pre-etching cavities on a device wafer, bonding the device wafer to a substrate wafer to form a substrate pair and aligning and bonding the cap wafer to the substrate pair to form a wafer stack with aligned cavities including a cap cavity and a device cavity. The wafer stack may be glassblown to form a dual-shell structure.

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Appl. No.: 18/347875

Filed: July 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240125599 A1	Apr. 18, 2024

Related U.S. Application Data

continuation parent-doc US 16836387 20200331 US 11703330 child-doc US 18347875
us-provisional-application US 62908459 20190930

Publication Classification

Int. Cl.: **G01C19/5719** (20120101); **B81B3/00** (20060101); **B81B7/00** (20060101); **B81C1/00** (20060101); **G01C19/5691** (20120101); **H03H3/007** (20060101); **H03H9/10** (20060101)

U.S. Cl.:

CPC **G01C19/5719** (20130101); **B81C1/00182** (20130101); **H03H3/0072** (20130101); **H03H9/1057** (20130101); B81B3/0051 (20130101); B81B7/0016 (20130101); B81B2201/0242 (20130101); B81B2201/0271 (20130101); B81B2203/0127 (20130101); B81B2203/0172 (20130101); B81C1/00158 (20130101); G01C19/5691 (20130101)

Field of Classification Search

CPC: G01C (19/5719); G01C (19/5691); B81C (1/00182); B81C (1/00158); H03H (3/0072); H03H (9/1057); B81B (3/0051); B81B (7/0016); B81B (2201/0242); B81B (2201/0271); B81B (2203/0127); B81B (2203/0172)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. patent application Ser. No. 16/836,387 titled FUSED QUARTZ DUAL SHELL RESONATOR AND METHOD OF FABRICATION filed on Mar. 31, 2020, U.S. Provisional Application No. 62/827,762 titled FUSED QUARTZ DUAL SHELL RESONATOR AND METHOD OF FABRICATION filed on Apr. 1, 2019 and U.S. Provisional Application No. 62/908,459 titled

FIELD

(1) The present disclosure generally relates to dual-shell resonator structures and methods of manufacturing.

BACKGROUND

(2) Implementation of Micro-Electro-Mechanical Systems (MEMS) gyroscopes in high-performance applications requires maintaining sensor integrity and survivability during high-spin and shock events. With conventional resonators, mechanical stresses induced in harsh dynamic environments can result in microcracks in high-stress concentration locations and fracture of sensor structural material. The displacement induced by the external shocks and vibrations during the operation of an electrostatically driven MEMS gyroscope could reduce the threshold of the dynamic pull-in stability and could result in snap-down of the resonant structure. Certain existing configurations with limited bonding surfaces may have weak points.

(3) Conventional structures exist for gyroscope protection including U.S. Pat. No. 9,429,428 to Skel et al. is directed to an Environmentally Robust Micro-Wineglass Gyroscope.

(4) There exists a need for improved structures and fabrication processes for resonator structures.

BRIEF SUMMARY OF THE EMBODIMENTS

(5) Disclosed and claimed herein are dual-shell structures, resonators and methods of fabrication.

(6) One embodiment is directed to a dual-shell resonator including an outer cap shell having an outer rim, and an inner device shell, wherein the outer cap shell and inner device shell are co-fabricated and share a common stem. The outer rim of the cap shell and a base of the inner stem are each configured to be fixed to a substrate layer and to provide an anchor for the inner device shell. The inner device shell includes a portion configured to vibrate within the outer cap shell. The outer cap shell, inner device shell and common stem are formed by triple stacked wafers with concentric cavities.

(7) In one embodiment, the outer cap shell is a hemispherical dome configured to house the inner device shell, and wherein the outer cap shell, inner device shell and stem are fused quartz.

(8) In one embodiment, the inner device shell is configured to operate using at least one of capacitive excitation and detection relative to the substrate layer.

(9) In one embodiment, the outer cap shell, the inner device shell and stem are co-fabricated by high-temperature glassblowing of the triple-stacked wafers, and wherein the common stem is a self-aligned stem.

(10) In one embodiment, a sensor is configured to measure vibrations of the inner device shell.

(11) In one embodiment, the outer cap layer includes a deposited piezo layer.

(12) In one embodiment, the outer cap layer includes at least one electrode configuration deposited by shadow mask.

(13) In one embodiment, the outer cap layer includes a differential driving configuration including at least a first set of electrode areas deposited on a first area and a second set of electrode areas deposited on a second area.

(14) In one embodiment, a first set of electrodes are deposited on a dome portion of the outer cap shell associated with the first area and the second set of electrodes are deposited in a recess of the cap shell associated with the second area, the first area perpendicular to the second area.

(15) In one embodiment, the dual shell resonator is configured to include a connection of the outer rim of the outer cap shell to a substrate, and a plurality of electrode connections coupled to electrode areas deposited on the outer cap shell.

(16) According to one embodiment, a method for fabricating a dual-shell resonator structure is provided. The method includes pre-etching cavities on a cap wafer, pre-etching cavities on a device wafer, bonding the device wafer to a substrate wafer to form a substrate pair, and aligning and

bonding the cap wafer to the substrate pair to form a wafer stack with aligned cavities including a cap cavity and a device cavity. The method also includes performing glassblowing of the wafer stack to form a dual-shell structure. The dual shell structure includes an outer cap shell having an outer rim, and an inner device shell, wherein the outer cap shell and inner device shell are co-fabricated and share a common stem. The outer rim of the cap shell and a base of the inner stem are each configured to be fixed to a substrate layer and to provide an anchor for the inner device shell. The inner device shell includes a portion configured to vibrate within the outer cap shell.

(17) In one embodiment, pre-etching cavities on the device wafer and cap wafer includes pre-etching fused quartz wafers by isotropic wet etching using Hydrofluoric (HF) acid.

(18) In one embodiment, the device wafer is bonded to the substrate wafer and the cap wafer is aligned and bonded to the substrate pair by plasma-assisted wafer bonding.

(19) In one embodiment, the outer cap shell is a hemispherical dome configured to house the inner device shell, and wherein the outer cap shell, inner device shell and stem are fused quartz.

(20) In one embodiment, the method also includes coating an outer surface of the outer shell with a piezo material layer.

(21) In one embodiment, the method also includes applying a thin-film metal conductive coating to an inner surface of the inner device shell and common stem.

(22) In one embodiment, the method also includes applying a thin-film metal bonding coating to a section of the outer cap shell rim.

(23) In one embodiment, the method also includes depositing a differential driving configuration on the outer cap layer, the differential driving configuration including at least a first set of electrode areas deposited on a first area and a second set of electrode areas deposited on a second area.

(24) In one embodiment, the method also includes releasing an inner shell by removing the substrate layer for integration of dual-shell resonator with a flat substrate for electrostatic actuation and detection.

(25) In one embodiment, the method also includes bonding the outer cap shell and common stem to a flat substrate by at least one of eutectic bonding, metal-metal compression bonding, and a low-outgassing conductive adhesives bonding, wherein a capacitive gap is provided between the inner device shell and the flat substrate.

(26) Other aspects, features, and techniques will be apparent to one skilled in the relevant art in view of the following detailed description of the embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features, objects, and advantages of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings in which like reference characters identify correspondingly throughout and wherein:

(2) FIG. 1 illustrates a perspective view of a dual-shell structure according to one or more embodiments;

(3) FIGS. 2A-2F show a graphical representation of a dual-shell structure fabrication according to one or more embodiments;

(4) FIG. 2G shows a graphical representation of a dual-shell structure according to one or more embodiments;

(5) FIGS. 3A-3G show a graphical representation of a dual-shell structure fabrication according to one or more embodiments;

(6) FIG. 3H is a graphical representation of an electrode substrate according to one or more embodiments;

(7) FIG. 3I is a graphical representation of an electrode substrate according to one or more

embodiments;

(8) FIGS. 4A-4C illustrates a dual-shell resonator according to one or more embodiments;

(9) FIG. 5 illustrates a process for fabricating a dual-shell resonator according to one or more embodiments;

(10) FIG. 6 illustrates a process for fabricating a substrate according to one or more embodiments;

(11) FIG. 7 is a graphic illustration of simulation results according to one or more embodiments;

(12) FIG. 8 is a graphic illustration of simulation results according to one or more embodiments;

(13) FIG. 9 is a graphic illustration of simulation results according to one or more embodiments;

(14) FIG. 10 is a graphic illustration of simulation results according to one or more embodiments;

(15) FIG. 11A illustrates a sectional view of a double-cavity triple stack according to one or more embodiments;

(16) FIG. 11B is a perspective view of a bonded wafer stack according to one or more embodiments;

(17) FIG. 12A illustrates a dual-shell structure having a metalized inner shell according to one or more embodiments;

(18) FIG. 12B illustrates a dual-shell die according to one or more embodiments;

(19) FIG. 13 illustrates frequency response according to one or more embodiments;

(20) FIG. 14A illustrates a frequency response for a dual-shell resonator according to one or more embodiments;

(21) FIG. 14B illustrates off-diagonal electrostatic tuning for a dual-shell resonator according to one or more embodiments;

(22) FIG. 14C illustrates frequency mismatch tuning for a dual-shell resonator according to one or more embodiments;

(23) FIG. 15 illustrates a sample sensor output for a gyro configuration according to one or more embodiments;

(24) FIGS. 16A-16F dual-shell resonator wafer according to one or more embodiments;

(25) FIG. 16F illustrates a cutaway representation of a dual-shell resonator wafer according to one or more embodiments;

(26) FIG. 17A illustrates a fabrication process according to one or more embodiments;

(27) FIG. 17B shows a graphical representation of a fabrication process according to one or more embodiments;

(28) FIG. 18 shows a shadow mask method according to one or more embodiments.

(29) FIGS. 19A-19B illustrate dual-shell resonator structures with electrodes according to one or more embodiments.

(30) FIG. 20 illustrates a simulated response of a dual-shell resonator according to one or more embodiments.

(31) FIG. 21 illustrates dual-shell resonator electrode configurations according to one or more embodiments.

(32) FIG. 22 illustrates dual-shell resonator electrode configurations according to one or more embodiments; and

(33) FIG. 23 illustrates response values of dual-shell resonator electrode configurations according to one or more embodiments.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(34) Overview and Terminology

(35) One aspect of the disclosure is directed to dual-shell resonator and resonator structures. Dual-shell structures are provided having an outer cap shell, inner device shell that are co-fabricates and share a common stem. The dual-shell structure may be configured to provide multiple connection points to a substrate. Another aspect of the disclosure is directed to processes and methods of fabricating dual-shell resonators.

(36) According to one embodiment, a dual-shell resonator is configured to provide an architecture

for a shell resonator gyroscope using a batch fabrication process. One or more embodiments are directed to an architecture and fabrication process based on wet-etching of cavities to device and cap wafers and bonding the two wafers by plasma bonding. The wet-etched features may be used for alignment of the wafers for plasma bonding. Embodiments described herein can be fabricated without requiring wafer bonding in a vacuum, and this reduce the complexity of manufacturing. In addition, structures and processes described herein parameters may be provided to minimize the displacement of a resonator when subjected to external high-g-shock and vibration. A dual-shell architecture as described herein can enable gyroscope operation through shock and vibration. In addition, the architecture can provide a co-fabricated cap shell to shield the gyroscope from environmental radiation, thermal fluctuation, airborne particles and as well as enabling in-situ vacuum sealing for an inner sensing element.

(37) According to one embodiment, dual-shell architecture processes are provided for fabrication of a three-dimensional fused quartz resonator. Fabrication can include defining two encapsulated concentric cavities using plasma-activated wafer bonding followed by high temperature glassblowing. The fabrication process can include use of triple wafer alignment and bonding using plasma bonding. Simulation results are provided herein. Structures described herein can be utilized as one or more of a resonator, gyroscope, and vibratory sensor in general.

(38) Another aspect of the disclosure relates to Micro-Electro-Mechanical Systems (MEMS) sensors, and more particularly to hemispherical resonator gyroscopes and for operation of the gyroscopes in a harsh environment. Configurations are provided to allow for 3D shell structures to be instrumented as the sensing element. According to one embodiment, fused quartz (FQ) shell resonators are provided for gyroscopes and fabricated by thermoplastic bulk deformation of FQ material at high temperatures using micro glassblowing or blow-torch molding. For gyroscope operation, the three-dimensional FQ shell resonators were assembled on a substrate for capacitive actuation and detection. In alternate embodiments, fused silica gyroscopes may be produced by using fused silica instead of fused quartz.

(39) According to one or more other embodiments, structures and fabrication processes are provided for piezo actuation of fused quartz dual shell resonators. Actuation of a resonator may include converting a signal (in most cases electric signal) into the motion of the resonator. For resonators made of electrically conductive material, such as silicon, the resonator structure itself can be used as an electric path for the signal. For resonators made of insulator, such as fused quartz, a metal coating may be necessary. Embodiments are directed to depositing and forming electrical paths and connections for one or more of delivering and receiving electric signals relative to the structure. Fused quartz may be characterized as having a very low intrinsic internal energy loss. Therefore, resonators made of fused quartz may provide a high mechanical quality factor compared to traditional silicon resonators. The noise level of a resonator may be inversely proportional to its mechanical quality factor. Therefore, a higher mechanical quality factor can lead to a lower noise level of a resonator, which is the goal of many applications. Embodiments are directed to fabrication processes for dual-shell structures and MEMS resonators.

(40) According to another embodiment, a structure and configurations are provided for a fused quartz dual-shell resonator that includes an inner shell as the resonant element, and the outer shell as the support and protection. According to another embodiment, piezo material deposited and shaped on the outer shell is utilized. As such, vibration of the outer shell can be transferred to the inner shell to actuate the resonant element. In certain embodiments, only the support of the resonator is coated with metal and piezo material, and the resonant part is kept intact. Therefore, the mechanical quality factor of the resonator due to actuation will not be reduced.

(41) According to one embodiment, a base electrode and a piezo material layer may be deposited on the outer cap layer, and the top electrode may be patterned and selectively deposited by shadow mask to actuate the device.

(42) As used herein, the terms “a” or “an” shall mean one or more than one. The term “plurality”

shall mean two or more than two. The term “another” is defined as a second or more. The terms “including” and/or “having” are open ended (e.g., comprising). The term “or” as used herein is to be interpreted as inclusive or meaning any one or any combination. Therefore, “A, B or C” means “any of the following: A; B; C; A and B; A and C; B and C; A, B and C”. An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.

(43) Reference throughout this document to “one embodiment,” “certain embodiments,” “an embodiment,” or similar term means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of such phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner on one or more embodiments without limitation.

Exemplary Embodiments

(44) FIG. 1 illustrates a perspective view of a dual-shell structure according to one or more embodiments. According to one embodiment, a dual-shell structure **100** is a resonator structure that may be employed for one or more sensing applications. A cutaway perspective view of dual-shell structure **100** is shown in FIG. 1. As will be described herein, dual-shell structure **100** is as a three-dimensional (3D) dual-shell structure that can be co-fabricated. In addition, dual-shell structure **100** may be formed by one or more glassblowing operations.

(45) According to one embodiment, dual-shell structure **100** includes an outer cap shell **110** and an inner device shell **120**. According to another embodiment, outer cap shell **110** and inner device shell **120** are co-fabricated by a micro glassblowing process and share a common inner stem **130**. Outer cap shell **110** may provide a housing for inner device shell **120** and provide an in-situ vacuum sealing capability. Inner stem **130** and the outer shell **110** may be bonded to a substrate at a stem attachment point **135** and at the outer shell rim **115**, respectively. Bonding dual-shell structure **100** can provide a fixed-fixed anchor for the resonator structure and for vibration of inner device shell **120** shell. According to another embodiment, a fixed-fixed anchor configuration of dual-shell structure **100** provides improved structural rigidity, high-g shock survivability, and allow for operation through shock and external vibrations. Inner device shell **120** may be configured to vibrate due to motion, acceleration, environmental changes, etc. of the structure. According to one embodiment, inner device shell **120** can operate to provide one or more degenerate wineglass modes using capacitive excitation and detection to perform the gyro operation.

(46) According to one embodiment, dual-shell structure **100** may be an oven-controlled sensor. By way of example, dual-shell structure **100** may be used with a temperature controlled (e.g., heated, etc.) isolation stage whose temperature is actively controlled so that sensor output is expected not to change when the surrounding temperature changes. The surface of the outer cap shell **110** can be coated with a low emissivity material for low power ovenization as well as radiation protection.

(47) One or more processes are described herein for co-fabrication of outer cap shell **110** and inner device shell **120** including high-temperature glassblowing of triple-stacked wafers with concentric cavities. According to certain embodiments, concurrent glassblowing forms dual-shell structure **100** such that post-fabrication alignment and bonding of the outer cap shell **110** to the device shell **120** is eliminated. According to one embodiment, dual-shell structure **100** is formed from fused quartz. Elements of the resonator may be co-fabricated such that the structure, while being discussed as having separate portions, that is a uniform structure. Dual-shell structure **100** may also include one or more additional materials or support the addition of materials by way of layering and deposits according to one or more embodiments.

(48) As mentioned above, dual-shell structure **100** is shown in a partial cutaway representation. According to one embodiment, dual-shell structure **100** may include a hemispherical outer cap shell **110** that fully encloses inner device shell **120** in its entirety (see FIG. 12A for example), the cutaway section in FIG. 1 is for illustration only of the structure.

(49) According to one embodiment, outer cap shell **110** includes dome shape having an outer surface and an inner surface **111**, the outer surface of outer cap shell **110** and inner surface **111** terminate into the outer shell rim **115**. According to one embodiment, outer shell rim **115** is configured to interface with a substrate to thus seal the inner portion of outer cap shell **110**. Outer shell rim **115** can provide support for the overall structure and the rim extends beyond dome shape to form a lip having a bottom surface for bonding.

(50) According to one embodiment, dome shape of outer cap shell **110** can include an indentation **131** that may provide conical cavity. Indentation **131** may be a recess in the central portion of the dome structure and outer surface of outer cap shell **110**, the bottom of indentation **131** may interface with common stem **130**, and in particular the top portion of common stem **130**.

(51) According to one embodiment, outer cap shell **110** can support one or more layers or deposits of material including piezo material and/or metal material. The one or more layers may be a piezo layer applied to a portion or the entire exterior dome surface of the outer cap shell. As will be described in more detail below, one or more electrode configurations may be provided to outer cap shell. According to one embodiment, the dome shape of outer cap shell **110** includes an inner surface and an outer surface as shown in FIG. 1.

(52) According to one embodiment, inner device shell **120** includes material that extends out from the inner stem with a curved shape, such as a hemispherical shape with the central portion co-fabricated with the common stem **130**. According to one embodiment, the connection point between inner device shell **120** and common stem **130** and the extension of inner device shell **120** matches or resembles the dome like architecture of the outer cap shell **110**. As such, inner device shell **120** curves down from its connection with common stem **130**. Inner device shell **120** can include a top surface **112** and inner surface. Inner device shell **120** may be separated from the inner surface **111** of outer cap shell **120**. Inner device shell **120** does not contact the substrate such that the rim/terminal edge **125** of inner device shell **120** does not include a lip and does not contact a substrate.

(53) According to one embodiment, common inner stem **130** relates to a central pillar of the resonator structure. Common inner stem **130** includes a stem attachment point **135** that relates to the bottom portion of the stem and that can be bonded to a substrate, whereas a top of common inner stem **130** is formed with outer cap shell **120**. According to one embodiment, inner device shell **120** extends from common inner stem **130** at a point separated from a connection of outer cap shell **110** with common inner stem **130**.

(54) According to one embodiment, dual-shell structure **100** may utilize fused quartz as the core sensing element for timing and inertial navigation applications. The structure of dual-shell structure **100** in a resonator can improve shock survivability and reduce vibration sensitivity by providing a fixed-fixed anchor for shell attachment, and stiffening the spurious tilt and out-of-plane modes of vibration.

(55) While dual-shell structure **100** is described in FIG. 1 as having certain elements and features, it should be appreciated that the embodiments described with respect to FIG. 1 may be applied to other embodiments described herein.

(56) FIGS. 2A-2F show a graphical representation of dual-shell structure fabrication according to one or more embodiments. One or more fabrication processes are provided that allow for formation and generation of a dual-shell structure (e.g., dual-shell structure **100** of FIG. 1). FIGS. 2A-2F provide graphical representations of one or more fabrication operations and may be employed to form a fused quartz dual-shell structure. One or more operations and processing operations of FIGS. 2A-2F may be employed to form a dual-shell structure including dual-shell resonator structures according to one or more embodiments.

(57) Embodiments described herein for fabrication of dual-shell structures **100** may be based on wafer bonding and high-temperature glassblowing processes. Fabrication may include pre-etching cavities on a cap fused quartz (FQ) wafer **210** and a device FQ wafer **220**. Pre-etching may be

based on and/or include an isotropic wet etching process using Hydrofluoric (HF) acid (48 wt. % in water solution). FIG. 2A is a side sectional view of a device wafer **220** cavity etch for the exemplary dual-shell structure. FIG. 2B is a side section view of a cap wafer **210** cavity etch for the exemplary dual-shell structure.

(58) FIG. 2C is a side section view of a triple wafer bonding for a dual-shell structure, wherein the triple wafer bonding is a substrate/device/cap triple stack plasma-assisted wafer bonding. The device wafer **220** may be bonded to the substrate wafer **210** to form a substrate pair. For example, in FIG. 2C, device wafer **220** is bonded to a blank FQ substrate wafer **240**, and the cap wafer **210** is aligned and bonded to the substrate/device pair **220/240** using plasma-assisted wafer bonding. The substrate/device/cap wafer stack **200** includes an array of aligned cavities including a cap cavity **217** in the device wafer **220** and device cavity **227** in cap wafer **210**. According to one embodiment, cap wafer **210** is bonded and aligned to the substrate pair to form a wafer stack with aligned cavities including a cap cavity and a device cavity. According to one embodiment, an outer cap shell, inner device shell and common stem are formed by triple stacked wafers with concentric cavities, such as device wafer **220** bonded to a blank FQ substrate wafer **240**, and the cap wafer **210** is aligned and bonded to the substrate/device pair **220/240**.

(59) FIG. 2D is a side section view of the exemplary dual-shell structure undergoing high-temperature glassblowing. Glassblowing of the wafer stack is performed to form a dual-shell structure, the glassblowing is performed at temperatures greater than 1500° C. At the glassblowing temperature, wherein the glassblowing temperature is set to a value greater than 1500° C., the pressure builds up simultaneously inside the two cavities **217** and **227**, and viscosity of FQ layers drops. The encapsulated cavities **217** and **227** expand and create a 3D dual-shell structure (e.g., dual shell structure **100**) with a self-centered stem (e.g., common inner stem **130**). During the viscous deformation of FQ, the surface tension would minimize the surface area and form a hemi-toroidal (or hemispherical) geometry.

(60) FIG. 2E is a side section view of a dual-shell structure, wherein the substrate has been removed and planarized by lapping. Outer cap shell **110** is a hemispherical dome configured to house the inner device shell **120** and the outer cap shell, inner device shell and stem are fused quartz. The inner device shell **120** was released by removing the substrate and polished using parallel back-lapping. The inner shell may be released by removing the substrate layer for integration of dual-shell resonator with a flat substrate for electrostatic actuation and detection.

(61) FIG. 2F is a side section view of assembly of a dual-shell structure on a substrate with through wafer vias, wherein the dual-shell structure and the substrate are joined by means of eutectic bonding in a vacuum furnace. According to one or more embodiments, bonding of the outer cap shell and common stem to a flat substrate can include one or more of eutectic bonding, metal-metal compression bonding, and a low-outgassing conductive adhesives bonding, wherein a capacitive gap is provided between the inner device shell and the flat substrate. The outer shell rim **115**, inner shell rim **125**, and the inner stem **130** were planarized in the lapping step and formed a reference plane for subsequent integration of the dual-shell resonator **100** with a flat substrate **300** for electrostatic actuation and detection (see FIGS. 4A-4C). In one embodiment, a thin-film metal conductive coating **228** is then applied to the inner surface of the inner shell **120** and the surface of a section of the inner stem **130**. In one embodiment, a thin-film metal bonding coating **229** is similarly applied to a section of the outer shell rim **115**. In one embodiment, the thin-film metal coatings **228** and **229** may include one or more of chromium (Cr), gold (Au), iridium (Ir), titanium nitride (TiN), and platinum (Pt).

(62) FIG. 2G shows a graphical representation of a dual-shell structure according to one or more embodiments. A bottom perspective view is shown in FIG. 2G of a fabricated prototype of a fused quartz dual-shell structure of FIG. 2F. In one embodiment, the outer cap shell **110** and inner stem **130** may be bonded to a flat substrate by means of eutectic bonding, metal-metal compression bonding, or a low-outgassing conductive adhesives bonding at points **140**. Once the outer shell **110**

and the inner stem **130** are bonded to the flat substrate, the inner shell rim **125** may have a capacitive gap remaining between it and a flat substrate. Adding a thin-film metal (such as thin-film metal **228**) as a conductive coating provides fused quartz insulator material with conductive coating to provide the electrical conductivity for electrostatic operation.

(63) FIGS. **3A-3G** show a graphical representation of dual-shell structure fabrication according to one or more embodiments. FIGS. **3A-3G** provide graphical representations of a substrate serving as a base for a fused quartz dual-shell structure. One or more operations and processing operations of FIGS. **3A-3G** may be employed to form a dual-shell structure (e.g., dual shell structure **100**) according to one or more embodiments. FIGS. **3H-3I** provide graphical representations of a dual-shell structure according to one or more embodiments.

(64) FIG. **3A** is a side section view of a cavity etch in a silicon wafer **310** to serve as the base of substrate **300**, which may be a flat substrate. The silicon wafer **310** may include a plurality of etched trenches **311** to serve as channels for glass reflow in a later step. FIG. **3B** is a side section view of a BSG wafer **320** bonded to the silicon wafer **310** in vacuum.

(65) FIG. **3C** is a side section view of the substrate **300**, wherein the substrate **300** is heated to above 900° C. and the glass **321** of the BSG wafer **320** has reflowed into the trenches **311** of the silicon wafer **310**. FIG. **3D** is a side section view of the substrate **300**, wherein the excess glass **321** has been removed and the surface of the substrate **300** has been planarized, thereby leaving planarized glass **322** in the etched trenches **311**.

(66) FIG. **3E** is a side section view of the substrate **300**, wherein a deep reactive-ion etching (DRIE) process is used to etch the backside of the silicon wafer **310** to define a set of capacitive gaps (such as gaps shown in FIGS. **4A-4C**). FIG. **3F** is a side section view of the substrate **300**, wherein a DRIE process is used again to etch the backside of the silicon wafer **310** to isolate the electrodes, thereby producing a pair of electrode isolation trenches **312**. Once a shallow etching **313** for the capacitive gaps and the isolation trenches **312** are produced, a silicon via **314** remains for the excitation and detection of vibration of an inner device shell (e.g., inner device shell **120**).

(67) FIG. **3G** is a side section view of the substrate **300**, wherein gold **315** is selectively deposited on the substrate **300** in preparation for the subsequent eutectic bonding step and getter material **316** is deposited on the substrate **300** to ensure long term vacuum stability after hermetic encapsulation of the dual-shell structure **100**. In one embodiment, once bonded, the gold **315** will bond the silicon wafer **310** to the thin-film metal bonding coating **229** applied to the outer shell rim **115** and the thin-film metal conductive coating **228** applied to the inner stem **130**. Coating on the outer shell edge may be bonded to gold **315** in the bonding area. The bonding can be performed using eutectic bonding method, or metal-metal compression bonding, or using a low-outgassing conductive adhesives bonding.

(68) FIG. **3H** is a top perspective view of an exemplary embodiment of a fabricated prototype of the electrode substrate **300** of FIG. **3G**. FIG. **3H** shows gold **315** bonding locations and silicon vias. FIG. **3I** is a bottom perspective view of a fabricated prototype of the electrode substrate **300** of FIG. **3G**. In one embodiment, the bottom side of the electrode substrate **300** includes a plurality of contact pads **317**, wherein each contact pad **317** is connected to a metal trace **318** configured to route signals. In one embodiment, each contact pad **317** and metal trace **318** pair is connected to a silicon via **314**.

(69) FIGS. **4A-4C** illustrate a dual-shell resonator according to one or more embodiments. According to one embodiment, dual-shell resonator **400** includes a dual-shell structure **100** joined to a substrate **300**. FIG. **4A** is a side section view of assembly of an exemplary embodiment of a dual-shell resonator **400** comprising a dual-shell structure **100** on a substrate **300** with through wafer vias **314**. According to one embodiment, dual-shell structure **100** and the substrate **300** are joined by means of eutectic bonding in a vacuum furnace. FIG. **4B** is a cutaway isometric view of the dual-shell resonator **400** of FIG. **4A**. FIG. **4C** is a perspective view of a fabricated prototype of the dual-shell resonator **400** of FIG. **4A**. In one embodiment, the planarized glass **322** isolates the

vias **314** from the surrounding substrate material. In one embodiment, the outer shell **110** and inner stem **130** may be bonded to the substrate **300** by means of eutectic bonding at points **140**. Once the outer cap shell **110** and the inner stem **130** are bonded to the flat substrate **300**, the inner shell rim **125** may have a capacitive gap **401** remaining between it and the flat substrate **300**. Outer cap shell **110** may have a hemispherical dome configured to house the inner device shell **120**, and the outer cap shell **110**, inner device shell **120**, and stem **130** are fused quartz.

(70) According to one embodiment, the inner device shell **120** is configured to operate using at least one of capacitive excitation and detection relative to the substrate layer. A sensor may be configured to measure vibrations of the inner device shell. In one embodiment, the dual-shell resonator **400** is capable of sensing directional motion by detecting vibrations that lead the inner shell rim **125** to contact one of the vias **314**, thereby creating a conductive path along the contact pad **317** and metal trace **318** of the inner stem **130**, through the thin-film metal conductive coating **228**, and then along the contact pad **317** and metal trace **318** of the contacted via **314**. In one embodiment, each of the plurality of contact pads **317** may be bonded to an electronic circuit or ASIC to enable sensory functions. In one embodiment, structures and fabrication methods described herein may include applying a thin-film metal conductive coating to an inner surface of the inner device shell **120** and common stem **130**. Similarly, a thin-film metal bonding coating may be applied to a section of the outer cap shell rim.

(71) FIG. 5 illustrates a process for fabricating a dual-shell resonator according to one or more embodiments. According to one embodiment, process **500** may be employed for fabricating the dual-shell resonator **100** of FIGS. 2A-2F. In one embodiment, at block **510**, the resonator fabrication process **500** includes coating a hard mask onto fused quartz. Alternatively, block **510** may use fused silica wafers instead of fused quartz.

(72) In one embodiment, at block **520**, the resonator fabrication process **500** includes patterning the hard mask and defining an opening for fused quartz etching and alignment marks for the wafer bonding steps. At block **530**, the resonator fabrication process **500** etches cavities in the device wafer **220** and the cap wafer **210** (see FIGS. 2A and 2B). At block **540**, the resonator fabrication process **500** directly bonds the device wafer **220** to a substrate wafer **240**.

(73) In one embodiment, at block **550**, the resonator fabrication process **500** aligns and bonds the stacked device/substrate wafer **220/240** to the cap wafer **210** (see FIG. 2C). At block **560**, the resonator fabrication process **500** dices the wafer stack **200** and conducts a glassblowing operation in a high-temperature furnace operating at temperatures greater than 1500° C. (see FIG. 2D). In one embodiment, at block **570**, the resonator fabrication process **500** releases the inner shell **120** by removing the substrate **240** with a lapping and polishing operation (see FIG. 2E).

(74) In one embodiment, at block **580**, the resonator fabrication process **500** metallizes the dual-shell resonator **100** by using atomic layer deposition or an evaporation process (see FIG. 2F).

(75) In one embodiment, at block **590**, the resonator fabrication process **500** selectively deposits gold on the inner stem **130** and the flange area of the outer shell **115** to prepare the dual-shell resonator **100** for bonding and vacuum sealing.

(76) FIG. 6 illustrates a process for fabricating a substrate according to one or more embodiments. Process **600** may be employed for fabricating the substrate of FIGS. 3A-3I.

(77) In one embodiment, at block **610**, the substrate fabrication process **600** etches the front side of a silicon wafer **310** in order to define cavities **311** for glass reflow (see FIG. 3A). In one embodiment, at block **620**, the substrate fabrication process **600** bonds the silicon wafer **310** with a BSG wafer **320** in a vacuum chamber (see FIG. 3B). At block **630**, the substrate fabrication process **600** melts the BSG wafer **320** using a high temperature furnace set to a temperature above 900° C. This causes the material of the BSG wafer **320** to flow into the cavities **311** of the silicon wafer **310**, thereby producing the glass reflow **321** (see FIG. 3C).

(78) In one embodiment, at block **640**, the substrate fabrication process **600** removes the excess glass reflow **321** and planarizes the silicon wafer **310**, thereby producing planarized glass **322** (see

FIG. 3D). In one embodiment, the step performed at block **640** may additionally planarize the substrate **300** as a whole, including the silicon wafer **310**, to produce a uniform plane on the front side of the substrate **300**.

(79) In one embodiment, at block **650**, the substrate fabrication process **600** uses a DRIE process on the back side of the silicon wafer **310** to produce a pair of electrode isolation trenches **312** for isolating the electrodes of the final product (see FIG. 3E).

(80) In one embodiment, at block **660**, the substrate fabrication process **600** again uses a DRIE process on the back side of the silicon wafer **310** to create a shallow etching **313** on top of the silicon vias **314** to define the capacitor gaps (see FIG. 4A). In one embodiment, at block **670**, the substrate fabrication process **600** selectively deposits gold **315** on the silicon wafer **310** for the subsequent eutectic bonding step (see FIG. 3G). In one embodiment, at block **680**, the substrate fabrication process **600** bonds the dual-shell structure **100** produced by the resonator fabrication process **500** to the substrate **300** produced by the substrate fabrication process **600** by means of eutectic bonding in a vacuum furnace (see FIGS. 4A-4C).

(81) FIG. 7 is a graphic illustration of simulation results according to one or more embodiments. Pressure diagram **700** illustrates pressure change in the cap cavity **711** and device cavity **706** in a generic glassblowing simulation, wherein the pressure drops as cavity volume increases and reaches an equilibrium with the ambient pressure. Diagram **700** shows cap cavity pressure **705** pressure over time compared to device cavity pressure **710**. FIG. 8 is a graphic illustration of an evolution of cap shell height (e.g., outer cap shell **110**) and device shell (e.g., inner device shell **120**) height in a generic glassblowing simulation, demonstrating the self-limiting nature of the process, wherein as cavity volume increases, the deformation rate reduces. Height diagram **800** illustrates of cap height **805** and device height **810**. The inset figure shows the deformation **811** due to blowing on the substrate side of the triple stack. The glassblowing may be a self-limiting process. The rate of deformation reduces over time until an equilibrium condition is reached and the 3D shell is fully developed. The final geometry of the glassblown shells (e.g., outer cap shell **110** and inner device shell **120**) may depend on initial geometry and process parameters. The volume of the pre-etched cavity, initial thickness, radius, glassblowing temperature, pressure, and wafer bonding pressure define the final geometry of a glassblown shell.

(82) The dual-shell glassblowing was simulated with a Newtonian isothermal fluid flow model with adaptive re-meshing in COMSOL Multi-physics FE Package. The thickness of the substrate, device, and cap wafer was 1 mm, 500 μm , and 500 μm , respectively, in all of the simulations. In this model, the instantaneous volume and pressure of the two cavities were calculated at each time step of the simulation. The initial volume of cavities was calculated based on the volume of the isotropic pre-etched cavities. It was assumed that wafers were bonded in atmospheric pressure and the initial pressure in the simulation was proportional to temperature rise. FIG. 7 demonstrates the pressure change in the cap cavity and device cavity during the simulation. As the pressure drops, the rate of deformation is reduced. The shells of a structure deform until the pressure reaches an equilibrium and limits the glassblowing height, FIG. 8. During the glassblowing, the bottom substrate would deform and change the geometry of the final structure. The larger backside blowing, the shell height would become smaller. In general, a thicker substrate wafer reduces the unwanted backside blowing.

(83) FIG. 9 is a graphic illustration of simulation results according to one or more embodiments. Simulation results **900** are based on parametric finite element results of the dual-shell glassblowing simulations at different cap/device thickness ratios, wherein the device thickness was 100 μm , the device radius was 4.5 mm, and the cap shell radius was 5 mm in all cases. FIG. 10 is a graphic illustration of a set of parametric finite element results of the dual-shell glassblowing simulations at different cap/device pressure ratios. For simulation result **1000** the cap and device thickness were 100 μm , the device radius was 4.5 mm, the cap shell radius was 5 mm, and the device cavity pressure was 1 atm in all cases.

(84) The dual-shell process was simulated for two cases. In the first case, the etch depth of cap wafer **210** was changed in a parametric FE simulation from 100 μm to 400 μm , and the etch depth of device wafer **220** was 400 μm . Thus, the thickness ratio of the cap layer **210** to the device layer **220** was varied from 4 to 1. FIG. **9** shows the predicted final geometry for four different thickness ratios based on height (y axis) compared to radius (x axis). The thinner the cap shell, the initial volume of the cap cavity would be larger, and subsequently, the cap shell would blow higher.

(85) In the second case, shown in FIG. **10**, the volume of the cap cavity **217** and device cavity **227** were constant. The initial pressure of the cap cavity **217** was varied to provide different cap-to-device pressure ratios, resembling the wafer bonding of the cap layer **210** at different pressures. The cap wafer **210** bonding at a lower pressure would reduce the blowing height of the cap shell **110**. The simulation results revealed that a variety of dual-shell geometries could be realized by controlling the pressure ratio and thickness ratio of the device wafer **220** and cap wafer **210**.

(86) FIG. **11A** illustrates a sectional view of a double-cavity triple stack according to one or more embodiments. FIG. **11B** is a perspective view of a bonded wafer stack according to one or more embodiments. In FIG. **11A**, a double-cavity triple stack **1100** (e.g., bonded stack **200**) is shown. FIG. **11B** is a perspective view of an exemplary bonded wafer stack **1101** (e.g., bonded stack **200**) showing the concentric cavities before glassblowing. High-purity FQ wafers with an initial thickness of 500 μm were used to fabricate the dual-shell structure. The wafers were coated with a 2 μm LPCVD doped polysilicon as a masking material for the isotropic wet etching. The device wafer **220** and the cap wafer **210** were time-etched in HF 48% to achieve 430 μm , and 250 μm etch depth, respectively. The device wafer was manually bonded to a 1 mm thick FQ substrate wafer **240**. The substrate/device wafer stack **240/220** and the cap wafer **210** were aligned and bonded in an AML-AWB aligner wafer bonder. The plasma-assisted direct bonding in atmospheric pressure was used for both wafer bonding steps. FIG. **11A** shows the cross-section of a triple-stack FQ die before the glassblowing. Fundamentally, the glassblowing process can be performed on the wafer scale using a high-temperature furnace with a uniform temperature distribution across the wafer. In the current fabrication process, the wafers were dies after the bonding step, and the glassblowing was performed on the die-level to operate the process in the uniform temperature zone of the furnace.

(87) FIG. **12A** illustrates a dual-shell structure having a metalized inner shell according to one or more embodiments. FIG. **12B** illustrates a dual-shell die according to one or more embodiments. In FIG. **12A** a glass-blown dual-shell structure **100** fabricated using triple-stack wafer bonding and high temperature glassblowing is shown the inner shell **120** was metalized for better visibility. FIG. **12B** is a perspective view of an exemplary released dual-shell die after substrate removal and planarization using back-lapping and polishing. According to one embodiment, a glassblowing process may be performed using a sliding tube Rapid Thermal Processing (RTP) furnace at 1800 K. The process time was between two to three minutes to fully develop a shell structure **100**. FIG. **12** illustrates a dual-shell structure **100** after the glassblowing. The substrate was removed using the parallel back-lapping to release the inner shell **120**. The rim of the inner shell **125** was polished using a 0.5 μm diamond abrasive polishing film.

(88) The frequency response of the released shell resonator prototype **100** was characterized using a bulk piezo stack for the excitation and a Laser Doppler Vibrometer (LDV) for detection. The cap shell **110** was attached to a bulk piezo stack, and the motion of the device shell **120** was captured from the rim area **125** using the LDV. FIG. **13** shows the frequency response of the inner shell **110** of the dual-shell structure **100** prototype, demonstrating the center frequency and the frequency mismatch at the $n=2$ and $n=3$ modes. The FE predictions were compared with measurements of the fabricated dual-shell structure **100** prototype. The final mesh of the glassblowing simulation was exported to a CAD software, the substrate region was trimmed, and the 3D dual-shell geometry was reconstructed for the modal analysis. A comparison between measurements and predictions are summarized in Table 1.

(89) TABLE-US-00001 TABLE 1 COMPARISON OF FE PREDICTION AND MEASUREMENTS ON THE FABRICATED DUAL-SHELL PROTOTYPE. Parameter FE Simulation Measurement Rim thickness (um) 86.2 86.2 Stem radius (mm) 0.32 0.48 Inner shell radius (mm) 3.33 3.24 Outer shell radius (mm) 4.06 3.9 Total dual-shell height (mm) 2.57 2.67 Resonant frequency (kHz) 8.7 9.6

(90) Table 1 illustrates frequency response according to one or more embodiments, including initial frequency response characterization of an exemplary embodiment of a dual-shell prototype using a bulk piezo stack excitation and LDV for detection, wherein no coupling was observed between the inner and outer shell. Table 1 provides a tabular comparison of FE prediction and measurements on the fabricated dual-shell prototype.

(91) Table 2 illustrates including sample results of frequency, frequency mismatch, and Q-factor. Table includes sample results of frequency, frequency mismatch, and Q-factor at the n=2 and n=3 modes.

(92) TABLE-US-00002 TABLE 2 n = 2 mode n = 3 mode Prototype f (kHz) df (Hz) Q-factor f (kHz) df (Hz) Q-factor #A1 9.671 47.9 262,530 23.263 68.8 1,321,700 #B1 4.905 26.5 945,607 14.581 4.4 1,169,392 #B2 5.499 65.1 1,129,969 16.223 87.6 1,237,700 #B3 4.782 30.9 721,161 13.953 12.7 797,887 #C1 7.914 8.2 313,600 15.638 15.1 1,613,136 #C2 5.456 65.7 1,252,811 15.836 61.6 1,533,332 #C3 6.261 68.2 141,969 16.354 67 862,521 #C4 6.958 30.7 <50,000 19.306 45.1 1,540,475

(93) FIG. 14A is a graphic depiction of a sample electrostatic frequency sweep for a dual-shell resonator. Frequency response **1400** is shown for a resonator without electrostatic tuning. FIG. 14B is a graphic depiction of a sample off-diagonal electrostatic tuning for modal decoupling for a dual-shell resonator. Frequency response **1405** is shown for a resonator based on off-diagonal tuning. FIG. 14C is a graphic depiction of a sample diagonal electrostatic tuning for frequency mismatch tuning for a dual-shell resonator. Frequency response **1410** is shown for a resonator based on off-diagonal tuning.

(94) FIG. 15 illustrates a sample sensor output for a gyro configuration according to one or more embodiments. Sample sensor output **1550** is shown for a resonator implemented as a gyro to sinusoidal rotation inputs for the dual-shell structure.

(95) FIGS. 16A-16F illustrate a dual-shell resonator wafer according to one or more embodiments. FIG. 16A is an isometric view of a dual-shell structure wafer **810** including a plurality of dual-shell structures **100** manufactured on a wafer-scale. FIG. 16B is a side view of the dual-shell structure wafer **810** of FIG. 16A. FIG. 16C is a side view of a back-lapped dual-shell structure wafer **830**, wherein back-lapping is performed on the dual-shell structure wafer **810** to remove the substrate and release the inner shells **120**. FIG. 16D is an isometric view of an electrode substrate wafer **840** corresponding with the dual-shell structure wafer **830** of FIG. 16C. FIG. 16E is an isometric view of an assembled dual-shell resonator wafer **850** comprising the back-lapped dual-shell structure wafer **830** of FIG. 16C joined with the electrode substrate wafer **840** of FIG. 16D. FIG. 16F is a cutaway isometric view of a dual-shell resonator **100** in the plurality of dual-shell resonators **100** comprising the dual-shell resonator wafer **850** of FIG. 16E.

(96) Piezo Actuation of Fused Quartz Dual Shell Resonator

(97) Actuation of a resonator is to convert a signal (in most cases electric signal) into the motion of the resonator. For resonators made of electrically conductive material, such as silicon, the structure itself can be used as an electric path for the signal. On the other hand, for resonators made of insulator, such as fused quartz, metal coating is generally necessary so that the electric signal can be delivered to the structure. Fused quartz is one of the materials whose intrinsic internal energy loss is very low. Therefore, resonators made of fused quartz theoretically can achieve very high mechanical quality factor than traditional silicon resonators. The noise level of a resonator is inversely proportional to its mechanical quality factor. Therefore, a higher mechanical quality factor can lead to a lower noise level of a resonator, which is the goal of many applications.

(98) However, it has been demonstrated that with metal coating as thin as 2 nm on the surface of fused quartz resonator, the mechanical quality factor of the resonator will be reduced by 5 times due to surface loss and mismatch of coefficient of thermal expansion between different materials: T. Nagourney, et al. "Effect of metal annealing on the Q-factor of metal-coated fused silica micro shell resonators." 2015 IEEE International Symposium on Inertial Sensors and Systems (ISISS), 2015. As a result, one or more of an improved actuation method and fabrication process is needed to preserve the designed high quality factor of the fused quartz dual shell resonator.

(99) According to one or more embodiments, piezo material is deposited and shaped on the outer shell. Vibration of the outer shell can be transferred to the inner shell to actuate the resonant element. In this way, only the support of the resonator is coated with metal and piezo material, and the resonant part is kept intact. Therefore, the mechanical quality factor of the resonator due to actuation will not be reduced. One or more embodiments are directed to utilization of the piezo layer, structuring the piezo layer, and the corresponding actuation process. A base electrode and a piezo material layer may be deposited on the outer cap layer, and the top electrode may be patterned and selectively deposited by shadow mask to actuate the device.

(100) By depositing piezo layer and electrodes on the outer shell, which serves as a support for the resonator, the surface of the inner shell, which is the resonant element, will be protected from metal coating. Therefore, a much higher mechanical quality factor of the resonator can be achieved. As a result, the noise level of the resonator will be much lower. Besides, the energy needed to actuate the resonator will be also lower due to the high quality factor.

(101) FIG. 17A illustrates a fabrication process according to one or more embodiments. According to one embodiment, process 1700 may be performed to form one or more dual shell structures and dual shell resonators (e.g., dual-shell structure 100, dual shell resonator 400, etc.). Process 1700 may be initiated by forming a dual-shell structure at block 1705. Formation of the dual-shell structure may include formation of an outer cap shell and inner device shell co-fabricated with a common stem. At block 1710, an adhesion layer and bottom electrode may be deposited for the dual-shell structure. At block 1715, piezo material may be deposited to the dual-shell structure. By way of example, an outer surface of the outer shell may be coated with a piezo material layer. At block 1720, a top electrode may be deposited. At block 1725, process 1700 includes releasing the shell. At block 1730, the shell may be assembled to a substrate.

(102) FIG. 17B shows a graphical representation of a fabrication process according to one or more embodiments. In certain embodiments, process 1750 is a graphical representation of process 1700 of FIG. 17A. Process 1750 may be initiated based on a dual-shell structure 1755. According to one embodiment, dual-shell structure 1755 may be a glassblown dual-shell structure. One or more of the processes and structures herein may employ the process of FIG. 17B.

(103) Dual-shell structure 1755 may include an inner device shell 1751 and outer cap shell 1752 co-fabricated with common stem 1753. Dual-shell structure 1755 may also include substrate 1754. After dual-shell structure 1755 is fabricated by micro-glassblowing technique, the surface should be cleaned by standard RCA-1 solution or Piranha solution to remove any organic residue before subsequent deposition process. An adhesion layer 1756 and bottom electrode 1757 may be deposited to outer cap shell 1752. According to one embodiment, the deposition may include titanium films sputter deposited on the surface and then annealed in Oxygen to form the Titanium dioxide layer as the adhesion layer 1756. Thickness of adhesion layer 1756 may be on the order of 200 nm. Then, a highly textured platinum layer can be sputter deposited on the adhesion layer 1756 as bottom electrode 1757. The thickness of Platinum layer can be on the order of 100 nm. In certain embodiments a shadow mask process is used to selectively deposit metal on curved surface. The shadow mask process may deposit one or more electrode configurations to an outer cap layer and/or one or more materials already on the outer cap layer including but not limited to metal, adhesion material, etc.

(104) Piezo material 1761 may then be deposited to adhesion layer 1757 and bottom electrode

1757. A top electrode **1766** may then be deposited over piezo material **1761**. Next, a 1 μm PZT layer with a Zr/Ti ratio of 52/48 can be deposited using a chemical solution deposition method with a crystallization temperature of 700° C. However, other piezo-electrical materials can be used for this purpose. The following step is for selective top electrode coating by shadow mask. The top electrode **1766** may be a 100 nm Platinum layer deposited by evaporation or sputtering. Finally, the resonator can be released shown by **1770** by lapping away the substrate and assembled to a ceramic package for further operation. A detailed process of piezo stack fabrication on flat surface has been reported, for example, in: D. M. Potrepka, et al. "Pt/TiO₂ growth templates for enhanced PZT films and MEMS devices." MRS Online Proceedings Library Archive 1299 (2011). Traditionally on flat devices, the top electrode, the piezo layer, and the bottom electrode can be patterned by lift-off process or ion milling. This disclosure describes techniques configured for a curved surface of structures including use of a shadow mask method.

(105) Following shell release, shell assembly may be performed including a plurality of connections of one or more electrode regions. Connection **1771** may relate to a connection of outer cap shell **1752** rim to substrate **1775**. In addition, one or more electrodes may be coupled to top electrodes of outer cap shell **1752**, such as electrodes **1772** and **1773**. By way of example, dual-shell resonator may be configured to include a connection **1771** for the outer rim of the outer cap shell to a substrate, and a plurality of electrode connections **1772**, **1773**, and **177** coupled to electrode areas deposited on the outer cap shell.

(106) FIG. **18** illustrates a shadow mask method according to one or more embodiments. According to one embodiment a shadow mask process **1800** may be employed for selective metal deposition on a dual shell structure **1805**. FIG. **18** shows a dual-shell structure **1805**, shadow mask **1815**, and frame **1810**. Frame **1810** may be used to align shadow mask **1815** with the structure to be deposited. Shadow mask **1815** can be batch-fabricated by deep reactive-ion etching (DRIE) of prime silicon wafer. Different shapes of opening can be designed to optimize the shape of the top electrode. In certain embodiments, frame **1810** can be machined, for example, from aluminum or graphite to withstand the high temperature during deposition and also eliminate out-gassing.

(107) FIGS. **19A-19B** illustrate dual-shell resonator structures with electrodes according to one or more embodiments. FIG. **19A** illustrates dual-shell resonator structure **1900** including outer cap shell **1905** and electrodes **1910** and **1911**. According to one embodiment, electrodes **1910** and **1911** may be deposited on outer cap shell **1905**. According to another embodiment, electrodes **1910** and **1911** may be fabricated using a shadow mask process (e.g., shadow mask process **1800**).

According to one embodiment, a designed shape of top electrodes **1910** and **1911** may result in deposition areas. FIG. **19B** shows a fabricated dual-shell resonator structure **1915** including outer cap shell **1920** and the deposition result including electrodes **1919** and **1922**.

(108) FIG. **20** illustrates a simulated response of a dual-shell resonator according to one or more embodiments. Transduction of the vibration from the outer shell to the inner shell has been demonstrated numerically. A method called Rapid Analytical-FEA Technique (RAFT) is used to simulate the frequency response of the dual shell resonator. The details of the method are reported in: J. M. Puder, et al. "Rapid Harmonic Analysis of Piezoelectric MEMS Resonators." IEEE transactions on ultrasonics, ferroelectrics, and frequency control 65.6 (2018): 979-990. The general idea is first to apply the finite element analysis to analyze the information of different modes, such as resonant frequency, effective stiffness, and energy stored due to the piezoelectric effect, and then use the analytical model to combine the information of different modes to generate the frequency response of the resonator. It is worth noticing that in this model, the quality factor of each mode should be manually assigned to generate the frequency response. For the n=2 resonant mode, a conservative quality factor value of 1e5 was assumed for the demonstration. An AC driving voltage of 10V was assumed. For these parameters, the predicted maximum displacement of the resonator is on the order of 10 μm , which is large enough to be detected optically. The complete frequency response of the dual shell resonator is shown in the figure below. All the major modes (the first 5

modes) have been captured.

(109) FIG. **20** illustrates a Rapid Analytical-FEA Technique (RAFT) response **2000** for a dual-shell resonator as described here. Displacement of the dual-shell resonator is shown by **2005** relative to frequency range. According to one embodiment, the $n=2$ mode of displacement is related to an electrode shape for driving effect. According to one embodiment, one or more of shape, position, and electrode configurations may be formed/fabricated based on displacement characteristics. In addition, shape of a top electrode may be controlled to optimize to increase transduction rate so that lower driving voltage is needed to achieve large resonant displacement. The optimization technique is reported in: J. S. Pulskamp, et al. "Electrode-shaping for the excitation and detection of permitted arbitrary modes in arbitrary geometries in piezoelectric resonators." IEEE transactions on ultrasonics, ferroelectrics, and frequency control 59.5 (2012): 1043-1060. According to one embodiment, a top electrode configuration may be deposited on the surface of the resonator where the signs of the principal stress are the same. By taking advantage of the same sign of the principal stress, both the e_{31} and e_{32} piezoelectric stress constants both excite or inhibit the mode, and as a result, the driving effect of the piezo layer will reach its maximum. According to one embodiment, an optimized electrode shape for $n=2$ mode of displacement **2005** is shown in the figure above.

(110) FIG. **21** illustrates dual-shell resonator electrode configurations according to one or more embodiments. Dual-shell resonator **2100** includes outer cap shell **2105** and one or more exemplary top electrode configurations. As discussed below, each electrode area may relate to the footprint or deposition area of electrode material on a shell resonator. According to one embodiment, a top electrode configuration includes electrode areas **2110** and **2115** on the dome or top portion and electrode areas **2117** and **2118** in a conical recess (e.g., blue electrode areas). In one embodiment, an electrode configuration can include a first set of electrodes deposited on a dome portion of the outer cap shell associated with the first area and the second set of electrodes deposited in a recess of the cap shell associated with the second area, the first area perpendicular to the second area. According to one embodiment, electrode areas **2110** and **2115** are on opposite sides of the conical deformation, each electrode area has the same size and occupies about a quarter of the top dome portion. Electrode areas **2110** and **2115** may relate to a top and bottom position, while electrode areas **2117** and **2118** relate to a right and left position of the conical recess. Alternatively, a top electrode configuration may include electrode areas **2111** and **2112**, and **2116** and **2119** in a conical recess portion (e.g., red electrode areas). Electrode configurations can include electrode areas associated with the red area or blue area of FIG. **21**. Blue electrode areas may have a similar location as red area electrodes, however, as shown in FIG. **21**, blue area electrodes may have a different external form, such as a slightly narrower width with respect to dome electrodes. In the case of differential driving configuration, two sets of electrodes can be deposited on the red and blue area on the outer cap layer, respectively. According to one embodiment, a first set of electrodes is associated with electrode area **2110** and **2115** (e.g., first area or areas) and a second set of electrodes is associated with electrode areas **2117** and **2118** (e.g., second area or areas). By optimizing electrode shape, such as the coverage area, position and external form, vibrational transduction can increase.

(111) FIG. **22** illustrates dual-shell resonator electrode configurations according to one or more embodiments. According to one embodiment, one or more electrode configurations may be used for a dual-shell resonator.

(112) Electrode configuration **2205** includes electrode areas **2215** on the top dome portion of the shell **2210** extending down the dome portion and into a conical recess. Electrode configuration **2205** includes electrode areas on alternate sides (e.g., left and right) of shell **2210**.

(113) According to one embodiment, electrode configuration **2220** includes electrode areas **2230** on the top dome portion of the shell **2210** extending down the dome portion but not into a conical recess. Electrode areas **2230** do not extend as low as electrode areas **2215** in electrode configuration **2205**.

(114) According to one embodiment, electrode configuration **2235** includes electrode areas **2240** in the recess of shell **2210** and does not include electrode areas on the top dome portion. According to one embodiment, electrode configuration **2250** includes electrode areas **2265** on the top dome portion of the shell **2210** extending down the dome portion and electrode areas **2265** in the recess of shell **2210**.

(115) FIG. **23** illustrates simulated response values of dual-shell resonator electrode configurations according to one or more embodiments. Response representations are shown charting response amplitude to driving voltage for a resonator structure. Response **2305** relates to an electrode configuration having electrode areas on a dome and in a recess of the cap dome in an alternate configuration, such as electrode configuration **2250** of FIG. **22**. Response **2310** relates to an electrode configuration having electrode areas in a recess of the cap dome, such as electrode configuration **2235** of FIG. **22**. Response **2315** relates to an electrode configuration having electrode areas on a dome on alternate sides, such as electrode configuration **2220** of FIG. **22**. Response **2320** relates to an electrode configuration having electrode areas on a dome and in a recess of the cap dome, such as electrode configuration **2215** of FIG. **22**.

(116) While this disclosure has been particularly shown and described with references to exemplary embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the claimed embodiments

Claims

1. A dual-shell resonator comprising: an outer cap shell having an outer rim, the outer cap shell having an indentation providing a conical recess; and an inner device shell, wherein the outer cap shell and inner device shell are co-fabricated and share a common stem, wherein the outer rim of the cap shell and a base of the common stem are each configured to be fixed to a substrate layer and to provide an anchor for the inner device shell, and wherein the inner device shell includes a portion configured to vibrate within the outer cap shell.
2. The dual-shell resonator of claim 1, wherein the outer cap shell is a hemispherical dome configured to house the inner device shell, and wherein the indentation is a conical recess in a central part of the outer cap shell, wherein a bottom of the recess interfaces with the common stem.
3. The dual-shell resonator of claim 1, wherein the inner device shell is configured to operate using at least one of capacitive excitation and detection relative to the substrate layer, and wherein a sensor is configured to measure vibrations of the inner device shell.
4. The dual-shell resonator of claim 1, wherein the outer cap shell, the inner device shell and stem are co-fabricated by high-temperature glassblowing of triple-stacked wafers with concentric cavities, and wherein the common stem is a self-aligned stem.
5. The dual-shell resonator of claim 1, wherein the outer cap shell includes an electrode in the conical recess.
6. The dual-shell resonator of claim 1, wherein the outer cap shell includes an electrode configuration including a first set of electrodes on a dome portion and a second set of electrodes in the conical recess.
7. The dual-shell resonator of claim 6, wherein the first set of electrodes are on opposite sides of the conical recess and associated with a top and bottom position and the second set of electrodes are associated with a right and left position of the conical recess.
8. The dual-shell resonator of claim 1, wherein the dual-shell resonator is configured to sense directional motion.
9. The dual-shell resonator of claim 1, wherein the an inner surface of the inner device shell and common stem include a thin film coating.
10. The dual-shell resonator of claim 1, wherein the inner device shell extends out from the common stem with a hemispherical shape and a central portion of the inner device shell is co-

fabricated with the common stem.

11. A method for fabricating a dual-shell resonator structure, the method comprising: pre-etching cavities on a cap wafer; pre-etching cavities on a device wafer; bonding the device wafer to a substrate wafer to form a substrate pair; aligning and bonding the cap wafer to the substrate pair to form a wafer stack with aligned cavities including a cap cavity and a device cavity; performing glassblowing of the wafer stack to form a dual-shell structure, wherein the dual shell structure includes: an outer cap shell having an outer rim, the outer cap shell having an indentation providing a conical recess; and an inner device shell, wherein the outer cap shell and inner device shell are co-fabricated and share a common stem, wherein the outer rim of the cap shell and a base of the inner stem are each configured to be fixed to a substrate layer and to provide an anchor for the inner device shell, wherein the inner device shell includes a portion configured to vibrate within the outer cap shell.

12. The method of claim 11, wherein pre-etching cavities on the device wafer and cap wafer includes pre-etching fused quartz wafers by isotropic wet etching using Hydrofluoric (HF) acid.

13. The method of claim 11, wherein the device wafer is bonded to the substrate wafer and the cap wafer is aligned and bonded to the substrate pair by plasma-assisted wafer bonding.

14. The method of claim 11, wherein the outer cap shell is a hemispherical dome configured to house the inner device shell, and wherein the indentation is a conical recess in a central part of the outer cap shell, wherein a bottom of the recess interfaces with the common stem.

15. The method of claim 11, further comprising coating an outer surface of the outer cap shell with an electrode in the conical recess.

16. The method of claim 11, further comprising applying an electrode configuration to the outer cap shell including a first set of electrodes on a dome portion and a second set of electrodes in the conical recess.

17. The method of claim 6, wherein the first set of electrodes are on opposite sides of the conical recess and associated with a top and bottom position and the second set of electrodes are associated with a right and left position of the conical recess.

18. The method of claim 11, wherein the inner device shell extends out from the common stem with a hemispherical shape, a central portion of the inner device shell co-fabricated with the common stem.

19. The method of claim 11, further comprising releasing an inner device shell by removing the substrate layer for integration of dual-shell resonator with a flat substrate for electrostatic actuation and detection.

20. The method of claim 11, further comprising bonding the outer cap shell and common stem to a flat substrate by at least one of eutectic bonding, metal-metal compression bonding, and a low-outgassing conductive adhesives bonding, wherein a capacitive gap is provided between the inner device shell and the flat substrate.
